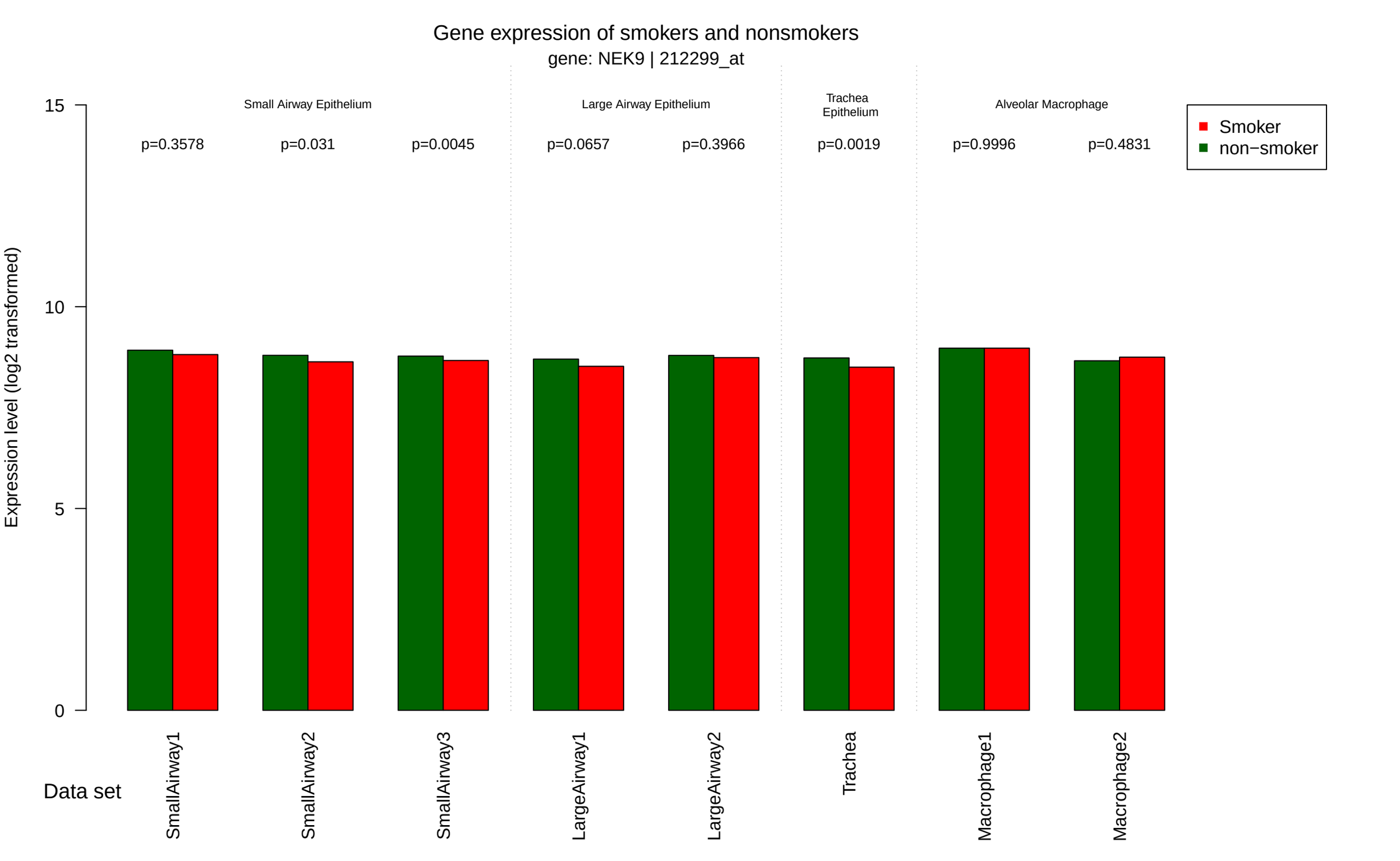
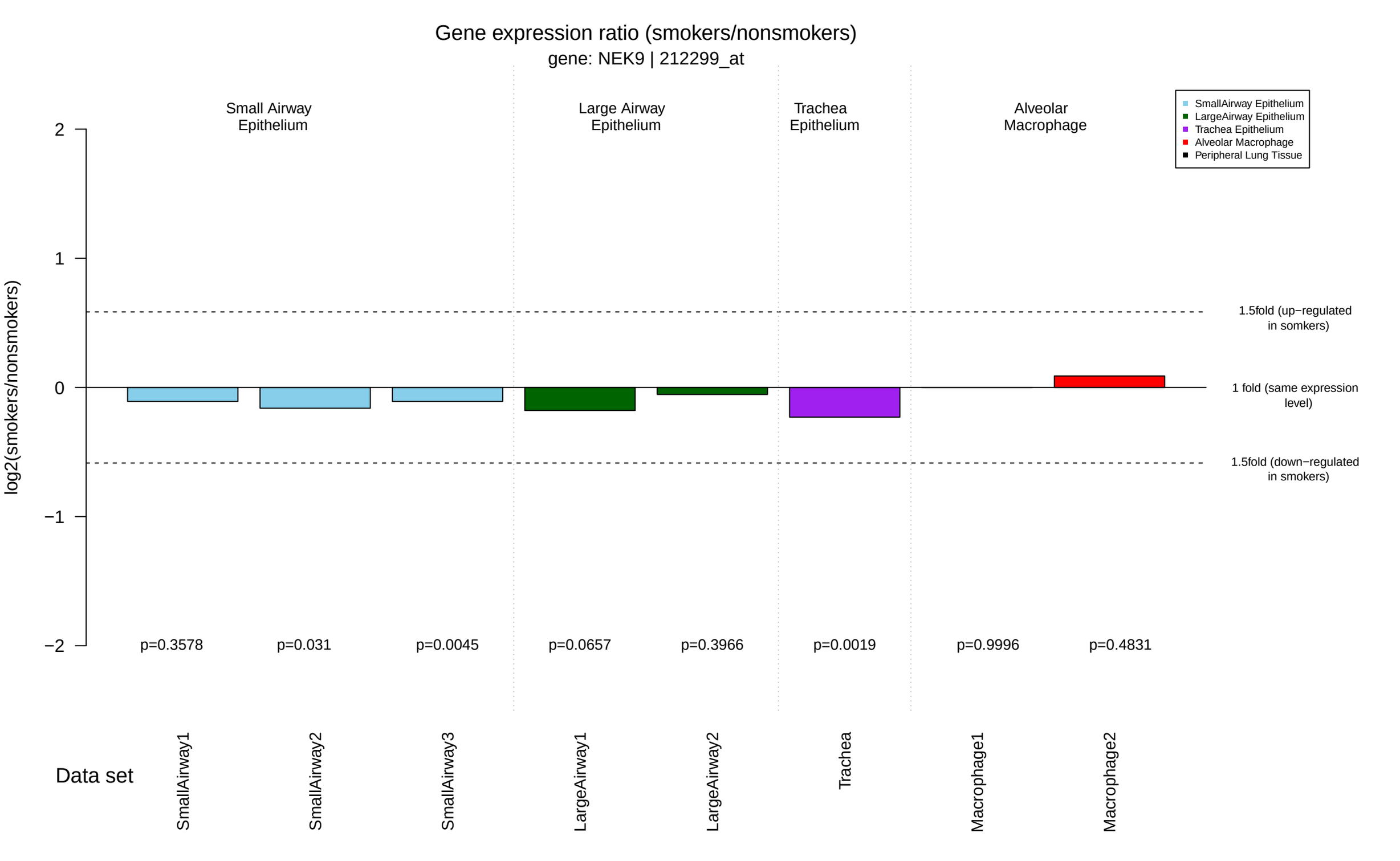
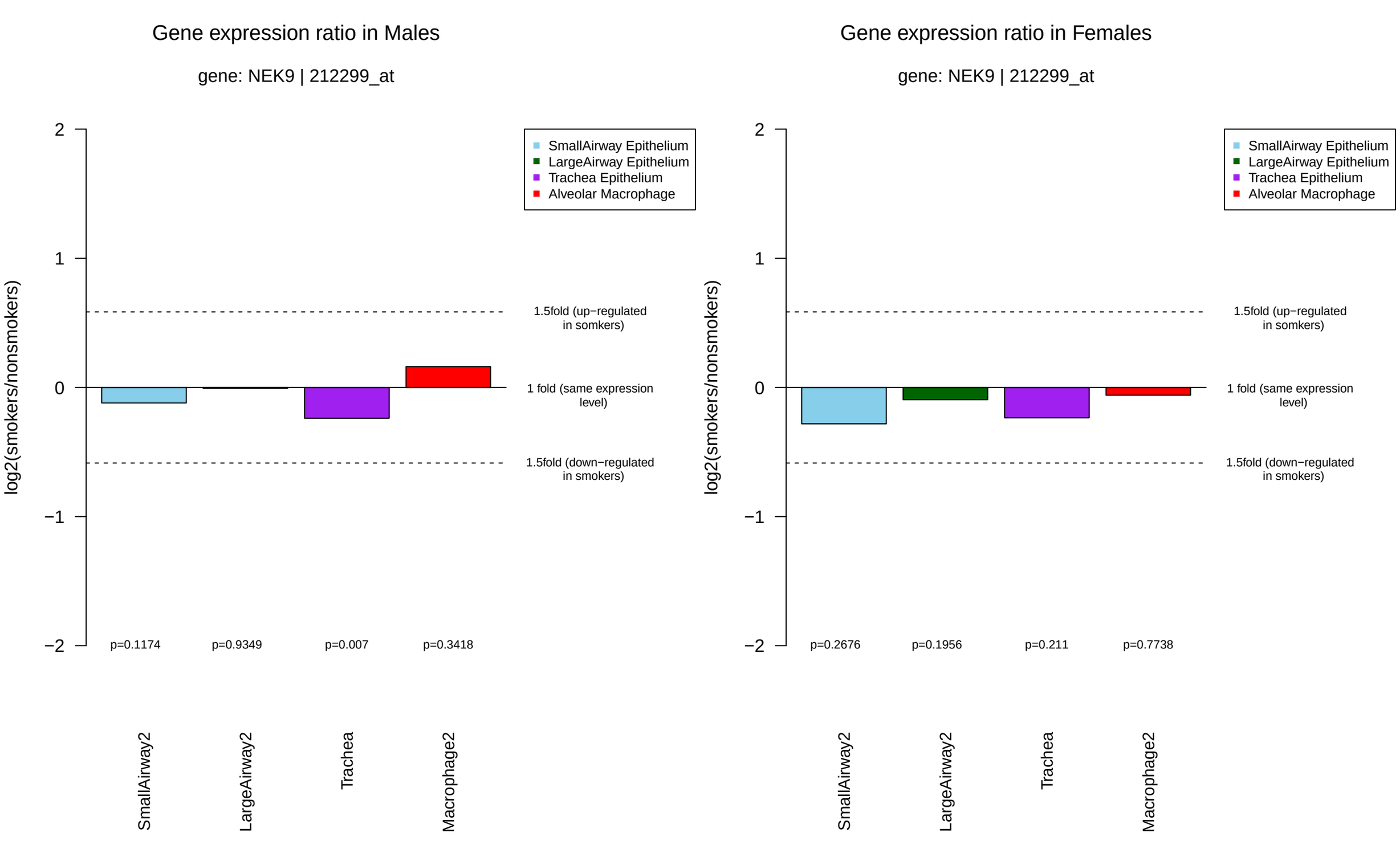
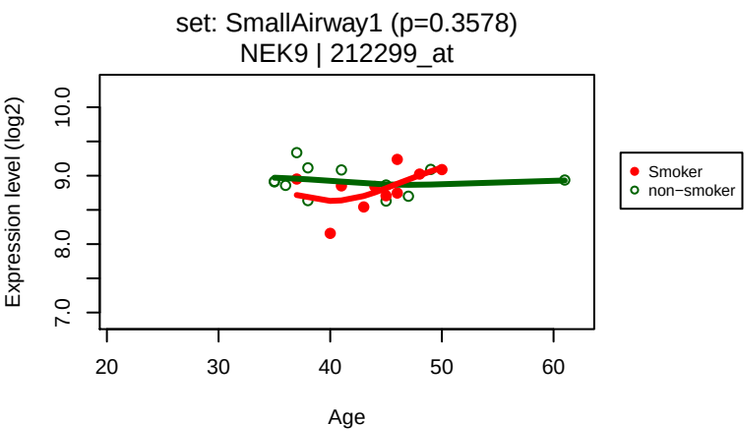
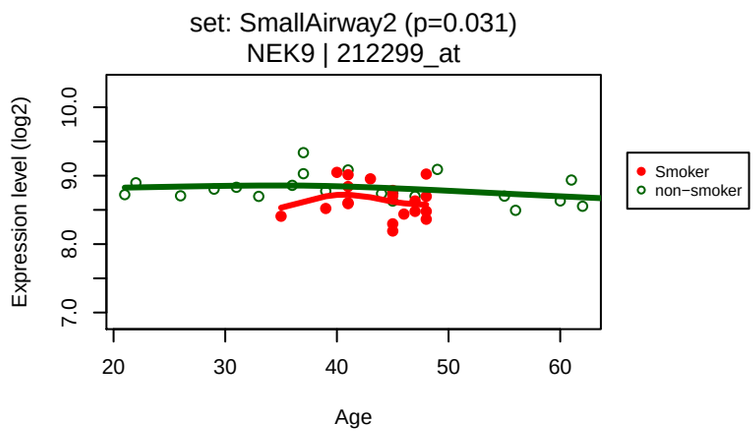
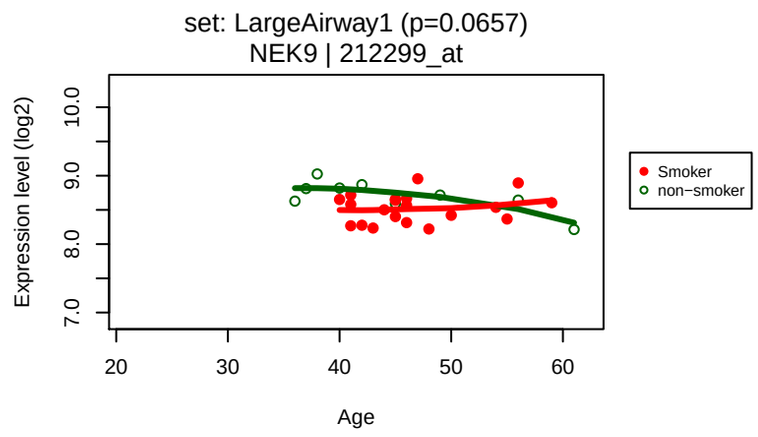

Gene expression of smokers and nonsmokers
gene: NEK9 | 212299_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3578
p=0.031
p=0.0045
p=0.0657
p=0.3966
p=0.0019
p=0.9996
p=0.4831
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

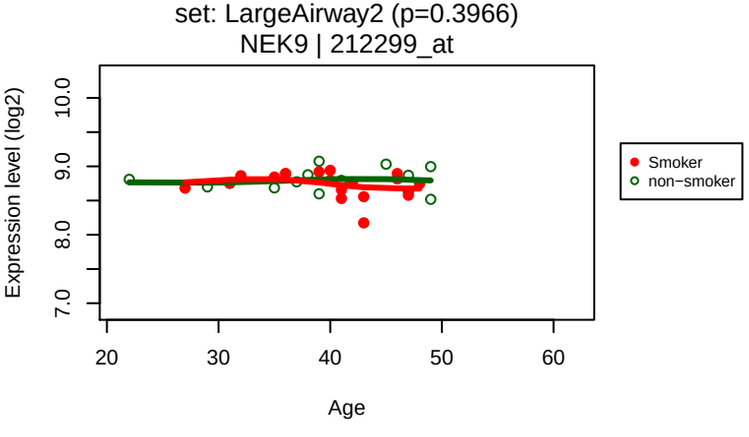

Gene expression ratio (smokers/nonsmokers)
gene: NEK9 | 212299_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3578
p=0.031
p=0.0045
p=0.0657
p=0.3966
p=0.0019
p=0.9996
p=0.4831
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

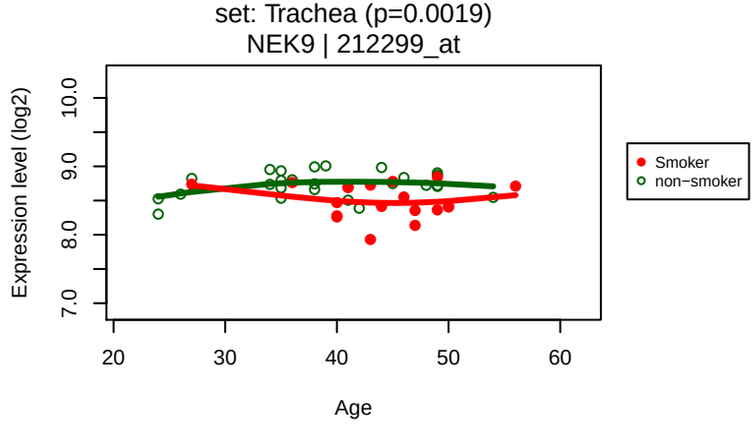

Gene expression ratio in Males
gene: NEK9 | 212299_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1174
p=0.9349
p=0.007
p=0.3418
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK9 | 212299_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2676
p=0.1956
p=0.211
p=0.7738
SmallAirway2
LargeAirway2
Trachea
Macrophage2

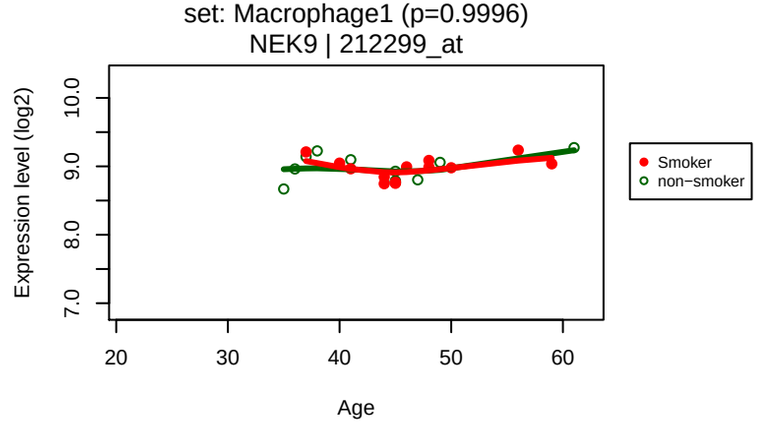

set: SmallAirway1 (p=0.3578)
NEK9 | 212299_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

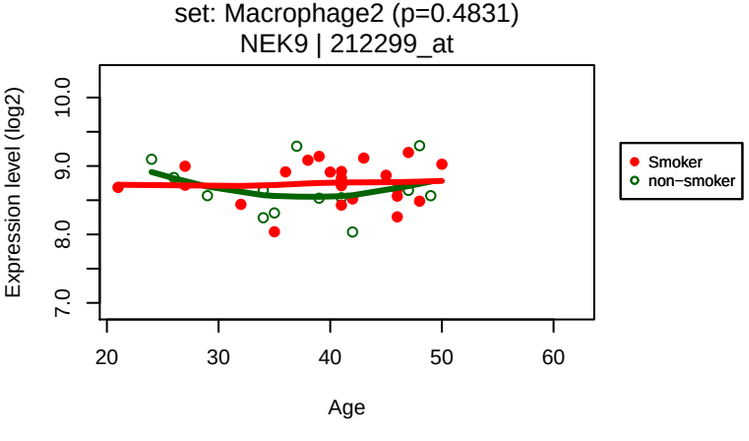

set: SmallAirway2 (p=0.031)
NEK9 | 212299_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

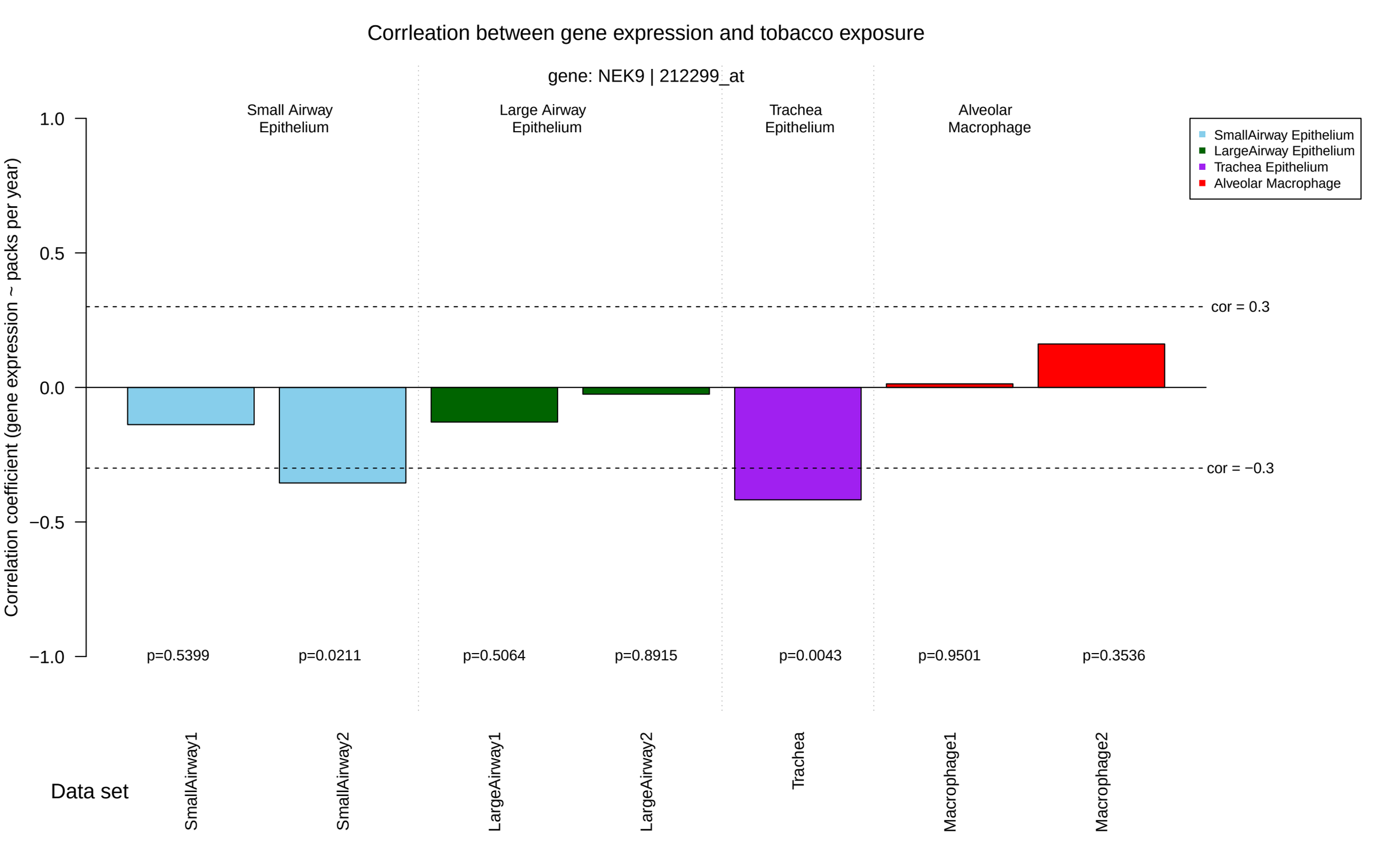

set: LargeAirway1 (p=0.0657)
NEK9 | 212299_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3966)
NEK9 | 212299_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0019)
NEK9 | 212299_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9996)
NEK9 | 212299_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4831)
NEK9 | 212299_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK9 | 212299_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5399
p=0.0211
p=0.5064
p=0.8915
p=0.0043
p=0.9501
p=0.3536
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2